CONCEPT 20.4

# The practical applications of DNA-based biotechnology affect our lives in many ways

In the last section, we'll survey the practical applications of DNA-based **biotechnology**, the manipulation of organisms or their components to make useful products. Today, major applications of DNA technology and genetic engineering include medicine, forensic evidence and genetic profiles, environmental cleanup, and agriculture.

## Medical Applications

One important use of DNA technology is the identification of human genes whose mutation plays a role in genetic diseases. These discoveries may lead to ways of diagnosing, treating, and even preventing such conditions. DNA technology has also identified genes that play a role in a number of "nongenetic" diseases, from arthritis to AIDS, by influencing susceptibility to these diseases. Furthermore, a wide variety of diseases involve changes in gene expression within the affected cells and often within the patient's immune system. By using RNA-seq and DNA microarray assays (see Figures 20.12 and 20.13) or other techniques to compare gene expression in healthy and diseased tissues, researchers are finding genes that are turned on or off in particular diseases. These genes and their products are potential targets for prevention or therapy.

### *Diagnosis and Treatment of Diseases*

A new chapter in the diagnosis of infectious diseases has been opened by DNA technology, in particular the use of PCR and labeled nucleic acid probes to track down pathogens. For example, because the sequence of the RNA genome of HIV is known, RT-PCR can be used to amplify and therefore detect and quantify even a small amount of HIV RNA in blood or tissue samples (see Figure 20.11). RT-PCR is often the best way to detect an otherwise elusive infectious agent.

Medical scientists can now diagnose hundreds of human genetic disorders by using PCR with primers that target the genes associated with these disorders. The amplified DNA product is then sequenced to reveal the presence or absence of the disease-causing mutation. Among the genes for human diseases targeted in this way are those for sickle-cell disease, hemophilia, cystic fibrosis, Huntington's disease, and Duchenne muscular dystrophy. Individuals with such diseases can often be identified before the onset of symptoms, even before birth (see Figure 14.19). PCR can also be used to identify symptomless carriers of potentially harmful recessive alleles as part of genetic counseling (see Concept 14.4).

### *Personal Genome Analysis*

As you learned earlier, genome-wide association studies have pinpointed SNPs (single nucleotide polymorphisms) that are linked to disease-associated alleles (see Figure 20.15). Individuals can be tested by PCR and sequencing for a SNP that is correlated with the abnormal allele. The presence of particular SNPs is correlated with increased risk for particular adverse health conditions such as heart disease, Alzheimer's disease, and some types of cancer.

Direct-to-consumer genome analysis companies offer kits allowing individuals to send in a swab containing cheek cells that the company will analyze genetically. Genetic testing for risk factors like heart disease, Alzheimer's disease, and some types of cancer is carried out by looking for linked SNPs that have been previously identified (see Figure 20.15). It may be helpful for individuals to learn about their health risks, with the understanding that such genetic tests merely reflect correlations and do not make predictions.

In addition to health-related genetic information, these companies compare a person's DNA segments with those from reference populations around the world, established from thousands of individuals of known ancestry. Based on how closely the segments match up, the report can tell an individual about their likely ancestral breakdown. Females can also learn about their maternal lines of descent, based on comparisons of mitochondrial DNA, which is contributed to the egg only by the mother. For males, an analysis of the sequence of the Y chromosome can trace their paternal lines of descent. As the size of the database used by these companies increases, the results become more refined and more accurate.

### *Personalized Medicine*

The techniques described in this chapter have also prompted improvements in disease treatments—for example, by knowing which specific cancer-related genes have been mutated in a particular patient's tumor. By analyzing the expression of many genes in large numbers of breast cancer patients, researchers can refine their understanding of the different subtypes of breast cancer (see Figure 18.27). Knowing the expression levels of particular genes in a given individual can help physicians determine the likelihood that the cancer will recur, thus helping them design an appropriate treatment. Given that some low-risk patients have a 96% survival rate over a ten-year period with no treatment, gene expression analysis allows doctors and patients access to valuable information when they are considering treatment options.

Many researchers envision a future of **personalized medicine**, a type of medical care in which each person's specific genetic profile can provide information about diseases or conditions for which the person is especially at risk